Recipe Data Cleaning

1- What major historical events happened in this time period in the United States that may have had an impact on culinary trends.

Here are a few major ones:

- **The Rise of Fast Food (1950s-present):** While starting earlier, the latter half of the 20th century saw the massive expansion and normalization of fast food, profoundly shaping American diets and preferences for convenience.
- **Increased Awareness of Health and Nutrition (1970s-present):** Concerns about heart disease, obesity, and other health issues led to greater interest in healthier eating, dietary guidelines, and the rise of diet foods.
- **The "Foodie" Movement and Celebrity Chefs (1980s-present):** A growing appreciation for gourmet cuisine, dining out, and the emergence of celebrity chefs like Julia Child and later Rachael Ray and Emeril Lagasse, popularized diverse cooking styles and fine dining.
- **Environmental and Ethical Concerns (1990s-present):** Movements advocating for organic, local, sustainable, and ethically sourced food gained traction, influencing consumer choices and the food industry.
- **Technological Advancements in Food Production and Distribution (throughout the period):** Innovations in agriculture, food processing, and logistics made a wider variety of foods available year-round and at lower costs, impacting both what and how people ate.

2- Given what we know about this data set, what kinds of information could we take from this data set that might be interesting to a person who is engaged in this topic?

The 'FAO.csv' dataset, which contains information on food and agriculture across various countries and years, a person interested in this topic could find a wealth of interesting insights. Here are some examples of what could be explored:

- **Global and Regional Food Security Trends:** Analyzing the 'Food supply (kcal/capita/day)' element across different 'Area's (countries/regions) over time to identify areas of improvement or concern regarding calorie availability.

- **Dietary Shifts:** Tracking the consumption trends of specific 'Item's (e.g., meat, grains, vegetables) to understand how diets have changed in different regions from 1961 to 2013.
- **Food Waste and Utilization:** Examining the 'Food' vs. 'Feed' elements for key agricultural products to understand how much is directly consumed by humans versus used for animal feed or other purposes.
- **Trade Dependency:** Investigating 'Import' and 'Export' data for various food items to assess a country's self-sufficiency or reliance on international trade for its food supply.
- **Impact of Geopolitical or Economic Events:** Correlating major global events (like droughts, conflicts, or economic crises) with changes in food production, supply, or trade in affected areas.
- **Agricultural Productivity:** Looking at 'Production' figures for different 'Item's in various 'Area's to observe changes in agricultural output over the decades.
- **Comparison of Developed vs. Developing Nations:** Grouping countries by economic status to compare their food supply trends, dietary patterns, and trade balances.
- **Specific Commodity Analysis:** Deep diving into a single commodity, like wheat or rice, to understand its global production, consumption, and trade dynamics.

3- Transforming the Dataset (Python Code) and download it as usa_feeding_categorized food.csv

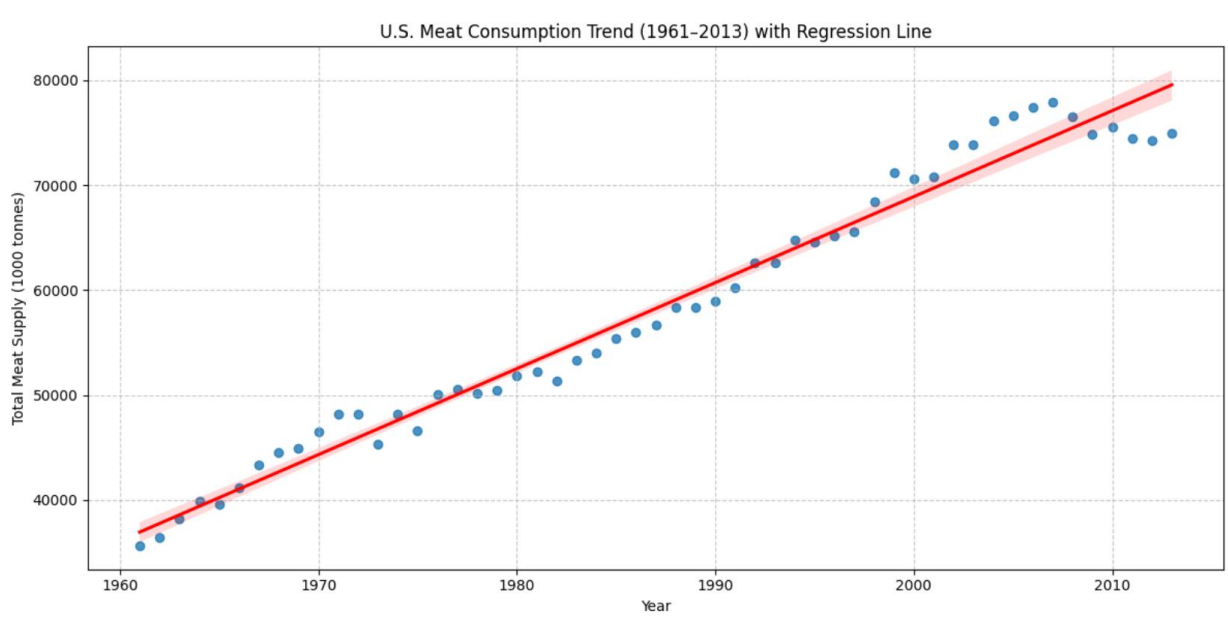
```
# Re-creating df_long from the original df to ensure it's populated
year_cols = [col for col in df.columns if col.startswith("Y")]
df_long = df.melt(
    id_vars=["Area Abbreviation", "Area Code", "Area",
             "Item Code", "Item",
             "Element Code", "Element",
             "Unit", "latitude", "longitude"],
    value_vars=year_cols,
    var_name="Year",
    value_name="Value"
)

# Clean Year column
df_long["Year"] = df_long["Year"].str.replace("Y", "").astype(int)

# Filter df_long for 'United States of America' data
df_long = df_long[df_long["Area"] == "United States of America"]

# Display the head of the filtered DataFrame
display(df_long.head())
```

	Area Abbreviation	Area Code	Area	Item Code	Item	Element Code	Element	Unit	latitude	longitude	Year	Value
20394	USA	231	United States of America	2511	Wheat and products	5521	Feed	1000 tonnes	37.09	-95.71	1961	1197.0
20395	USA	231	United States of America	2511	Wheat and products	5142	Food	1000 tonnes	37.09	-95.71	1961	13439.0
20396	USA	231	United States of America	2805	Rice (Milled Equivalent)	5142	Food	1000 tonnes	37.09	-95.71	1961	481.0
20397	USA	231	United States of America	2513	Barley and products	5521	Feed	1000 tonnes	37.09	-95.71	1961	5247.0
20398	USA	231	United States of America	2513	Barley and products	5142	Food	1000 tonnes	37.09	-95.71	1961	131.0

4- Questions to answer

Q1: Did meat consumption change across year?

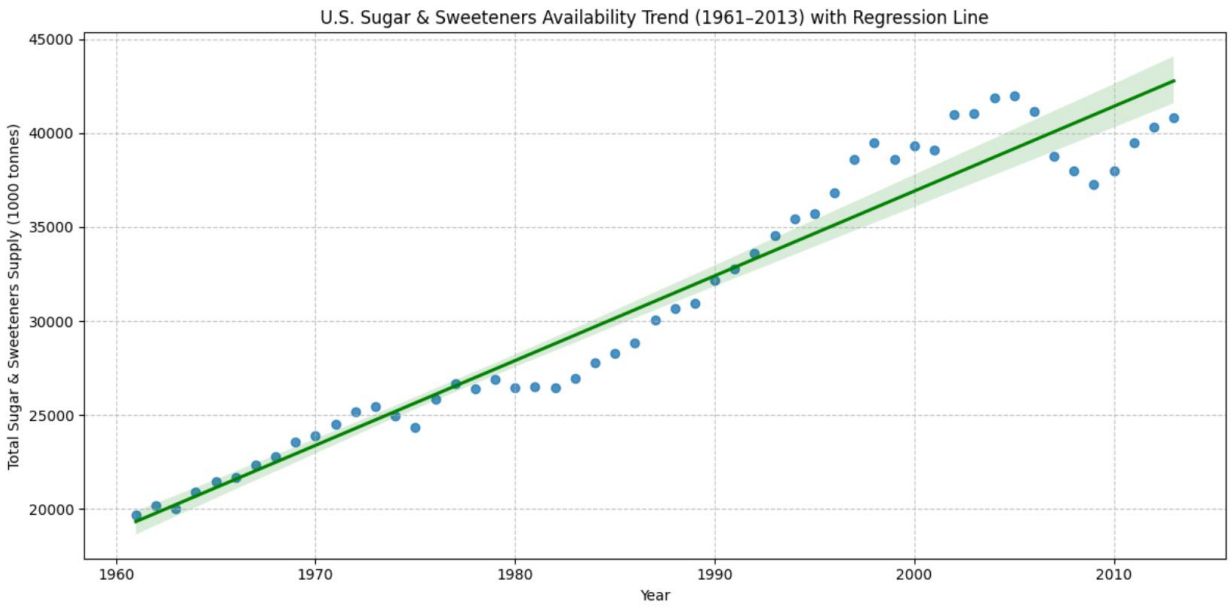
```
import statsmodels.api as sm
import matplotlib.pyplot as plt
import seaborn as sns

# Assuming df_us_long with 'FoodCategory' is already available

# 1. Filter for Meat category and 'Food' element (human consumption)
meat_consumption_data = df_us_long[
    (df_us_long['FoodCategory'] == 'Meat') &
    (df_us_long['Element'] == 'Food')
]

    # 2. Aggregate Total Meat Consumption per Year
meat_trend = meat_consumption_data.groupby("Year")["Value"].sum().reset_index()

# 3. Prepare data for linear regression
# Add a constant to the independent variable for the intercept term
X = sm.add_constant(meat_trend["Year"])
y = meat_trend["Value"]

# 4. Run Ordinary Least Squares (OLS) Regression
model = sm.OLS(y, X).fit()

# 7. Plot the trend with the regression line
plt.figure(figsize=(12,6))
sns.regplot(data=meat_trend, x="Year", y="Value", ci=95, line_kws={'color':'red'})
plt.title("U.S. Meat Consumption Trend (1961-2013) with Regression Line")
plt.xlabel("Year")
plt.ylabel("Total Meat Supply (1000 tonnes)")
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



Conclusion: U.S. meat consumption for direct human intake has experienced a substantial and statistically undeniable increase between 1961 and 2013.

Q2: Has sugar availability increased over time?

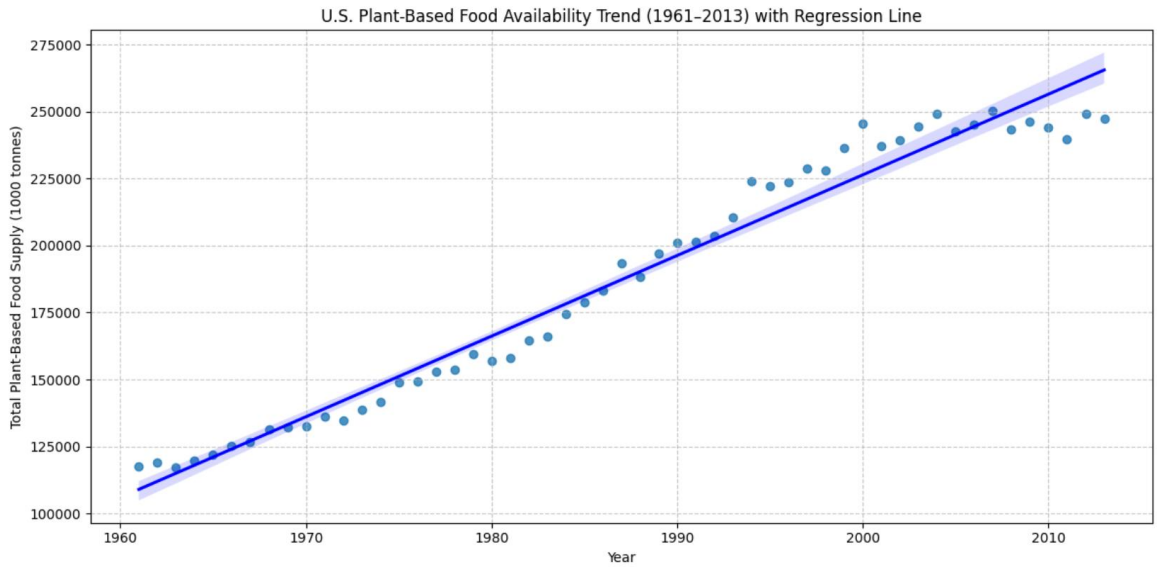
```
# 1. Filter for 'Sugar & Sweeteners' category and 'Food' element (human consumption)
sugar_consumption_data = df_us_long[
    (df_us_long['FoodCategory'] == 'Sugar & Sweeteners') &
    (df_us_long['Element'] == 'Food')
]

# 2. Aggregate Total Sugar Consumption per Year
sugar_trend = sugar_consumption_data.groupby("Year")["Value"].sum().reset_index()

# 3. Prepare data for linear regression
# Add a constant to the independent variable for the intercept term
X_sugar = sm.add_constant(sugar_trend["Year"])
y_sugar = sugar_trend["Value"]
# 7. Plot the trend with the regression line
plt.figure(figsize=(12,6))
sns.regplot(data=sugar_trend, x="Year", y="Value", ci=95, line_kws={'color':'green'})
plt.title("U.S. Sugar & Sweeteners Availability Trend (1961–2013) with Regression Line")
plt.xlabel("Year")
plt.ylabel("Total Sugar & Sweeteners Supply (1000 tonnes)")
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



Conclusion: the analysis clearly demonstrates a highly significant and consistent increase in U.S. sugar and sweeteners availability over the period from 1961 to 2013, with an average annual increase of 450.76 thousand tones.

Q3- Did plant-based foods increase during health-conscious across years?

```
# Define plant-based food categories from the food_categories dictionary
plant_based_categories = ['Vegetables', 'Fruits', 'Cereals', 'Oils & Fats']

# 1. Filter for plant-based food categories and 'Food' element (human consumption)
plant_based_consumption_data = df_us_long[
    (df_us_long['FoodCategory'].isin(plant_based_categories)) &
    (df_us_long['Element'] == 'Food')
]

# 2. Aggregate Total Plant-Based Food Consumption per Year
plant_based_trend = plant_based_consumption_data.groupby("Year")["Value"].sum().reset_index()

# Ensure plant_based_trend is not empty before proceeding
if plant_based_trend.empty:
    print("No plant-based food consumption data found for analysis after filtering.")
else:
    # 3. Prepare data for linear regression
    # Add a constant to the independent variable for the intercept term
    X_plant = sm.add_constant(plant_based_trend["Year"])
    y_plant = plant_based_trend["Value"]

    # 7. Plot the trend with the regression line
    plt.figure(figsize=(12,6))
    sns.regplot(data=plant_based_trend, x="Year", y="Value", ci=95, line_kws={'color':'blue'})
    plt.title("U.S. Plant-Based Food Availability Trend (1961–2013) with Regression Line")
    plt.xlabel("Year")
    plt.ylabel("Total Plant-Based Food Supply (1000 tonnes)")
    plt.grid(True, linestyle='--', alpha=0.7)
    plt.tight_layout()
    plt.show()
```



Conclusion: the visualization clearly demonstrates a highly significant and consistent increase in U.S. plant-based food availability over the period from 1961 to 2013, with an average annual increase of 3011.29 thousand tones. This suggests that during the era of increasing health consciousness, the supply of plant-based foods in the U.S. has indeed grown significantly.